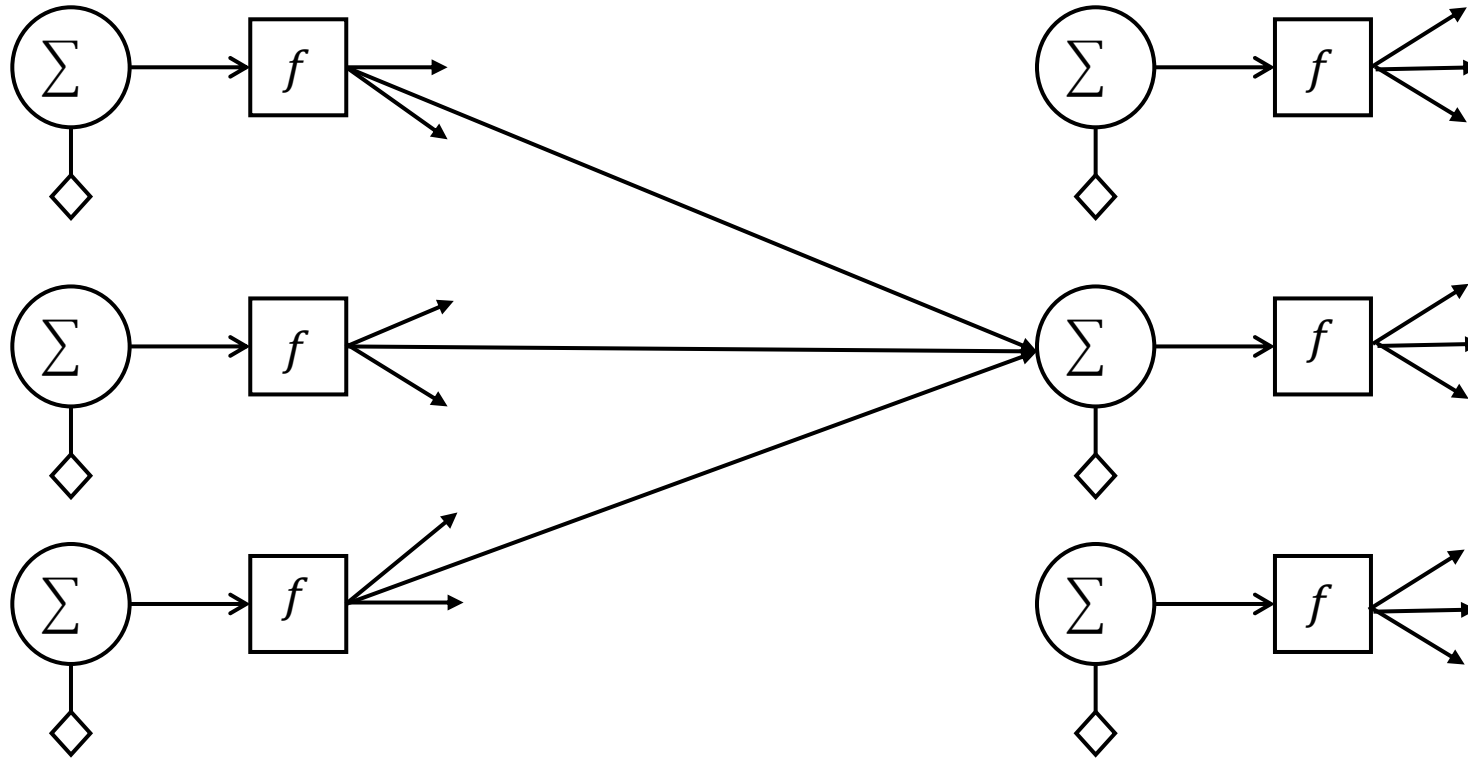
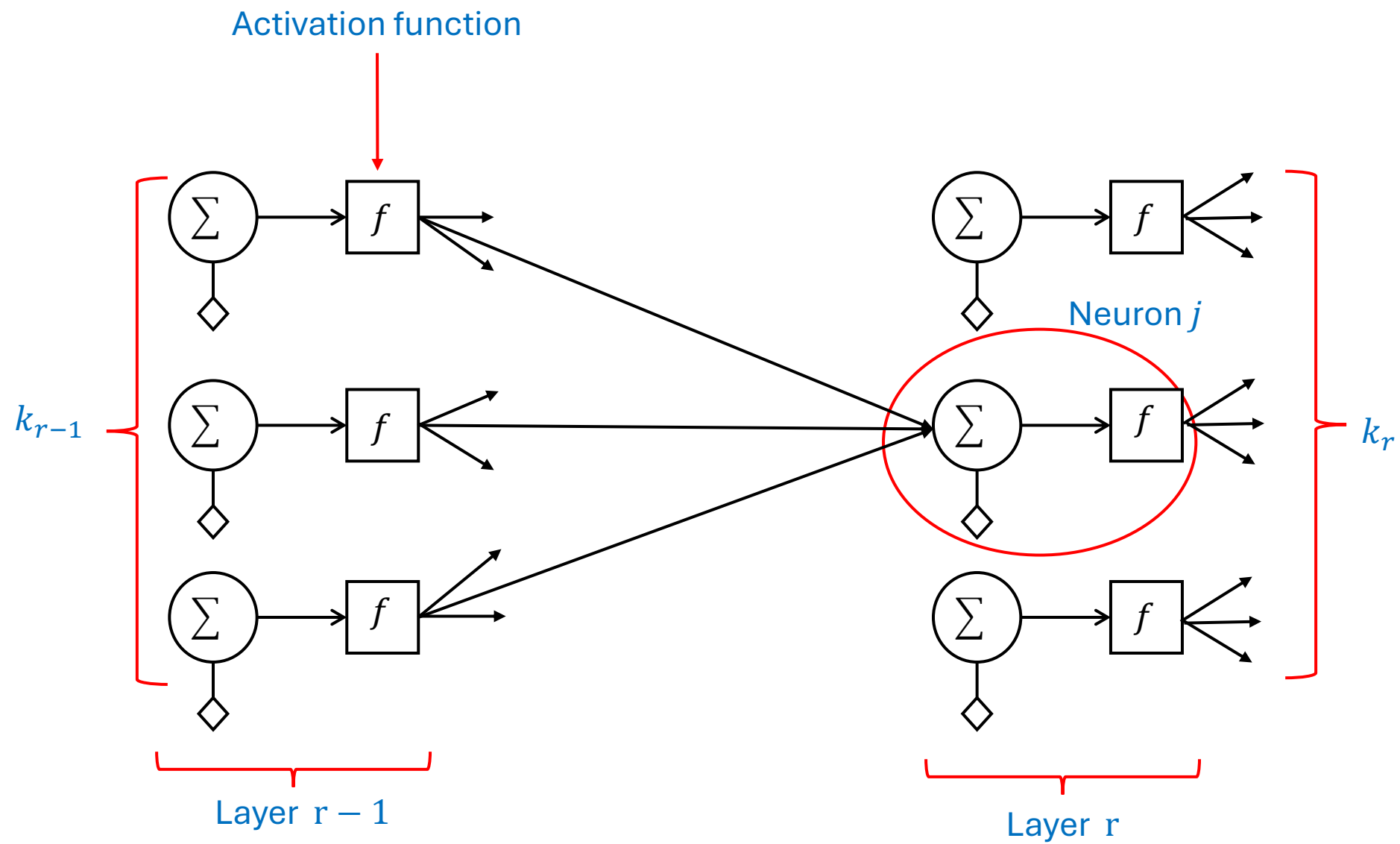


Multilayer Neural Network

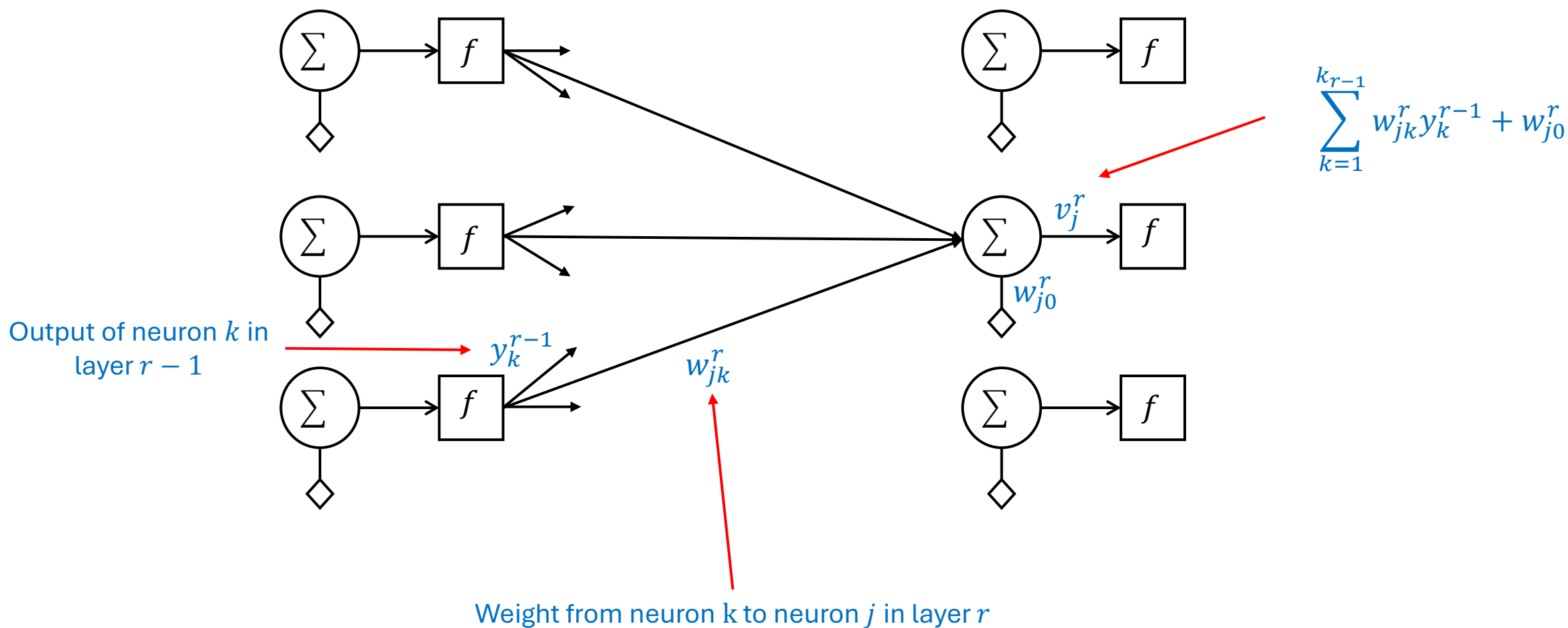


- L Number of layers
- $k_0 = l$ Number of neurons in the input layer
- k_r Number of neurons in the layer $r = 1, \dots, L$
- N Number of observations



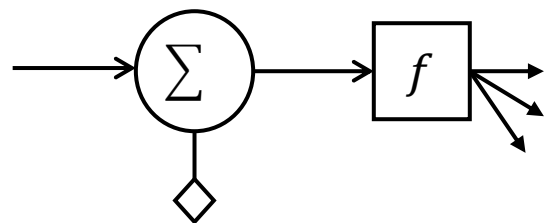
$$\mathbf{w}_j^r = [w_{j0}^r, w_{j1}^r, \dots, w_{jk}^r]^T$$

Weights of neuron j in layer r (including threshold)

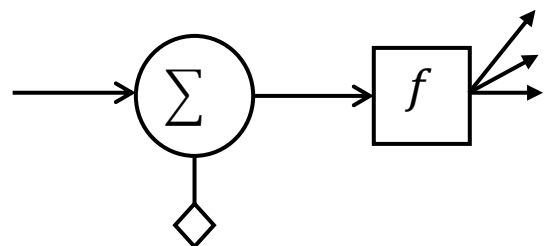
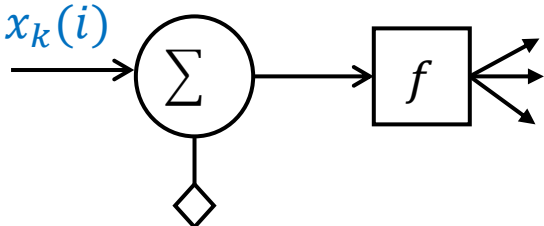


Network's
inputs

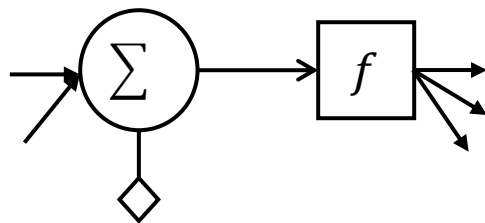
$x(i)$



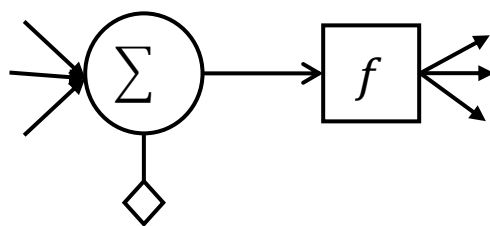
$y_j^0 = x_k(i)$



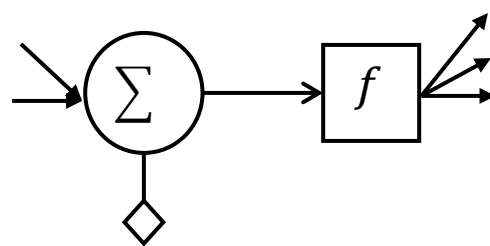
Hidden layers



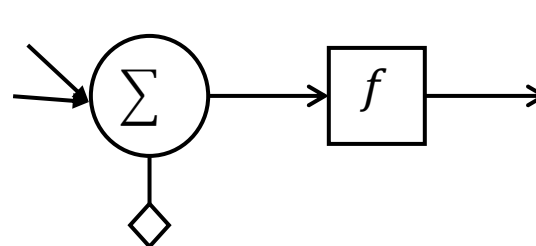
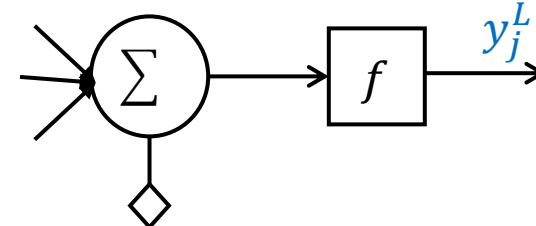
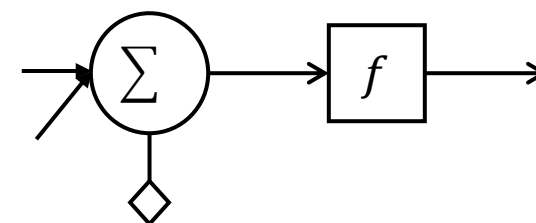
...



...



...



Network's
outputs

$\hat{y}(i)$



$y_j^L = \hat{y}_j(i)$